# WORKSHEET - NAPLAN Year 9

**Number and Algebra**

**Basic Calculations and Algebra (9)**

**Teacher:** Joanne Rust

**Exam Equivalent Time:** 7.5 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

## Questions

1. Number and Algebra, NAP-25088

Which one of the following expressions is equivalent to $4(3m-1)$?

$7m-1$ $\quad$ $12m-1$ $\quad$ $12m-4$ $\quad$ $8m$

2. Number and Algebra, NAP-48609

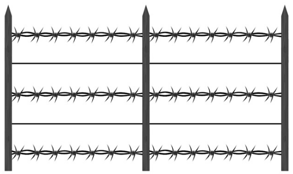
Which expression is equivalent to $5-6t$?

$6-5t$ $\quad$ $6t-5$ $\quad$ $-5+6t$ $\quad$ $-6t+5$

3. Number and Algebra, NAP-19173

Farmer Jim uses 5-strand wire fencing around his paddocks.

His fences have 3 barbed wire strands and 2 plain wire strands, as shown in the diagram below.



Barbed wire costs $\$b$ per metre. Plain wire costs $\$w$ per metre.

Which of these expressions gives the total cost of the wire needed for a fence of length $F$ metres?

$5bwF$ $\quad$ $6bwF$ $\quad$ $6(b+w)F$ $\quad$ $(3b+2w)F$

4. Number and Algebra, NAP-89736

Aaron has $3 less than Brendon.

Using $a$ for Aaron's money and $b$ for Brendon's money, which equation correctly describes this fact?

$a=b+3$ $\quad$ $a=3-b$ $\quad$ $a=b-3$ $\quad$ $a=b\div 3$

5. Number and Algebra, NAP-25117

In these expressions $m$ and $n$ are positive whole numbers and $l$ is a negative whole number.

Which expression gives the largest positive value?

- $(m+n)\times l$
- $(m-n)\div l$
- $(m-n)\div -l$
- $(m+n)\times -l$

6. Number and Algebra, NAP-42727

An insect expert estimates that an anthill 30 cm high contains $10^8$ ants.

He also estimates that a 25 cm high anthill contains $10^7$ ants.

If his estimations are correct, how many more ants live in the 30 cm high anthill than the 25 cm high one?

9 million $\quad$ 10 million $\quad$ 90 million $\quad$ 100 million

## Worked Solutions

1. Number and Algebra, NAP-25088

$$4(3m-1)$$
$$= (4 \times 3m) - (4 \times 1)$$
$$= 12m - 4$$

2. Number and Algebra, NAP-48609

$$-5 + 6t$$

3. Number and Algebra, NAP-19173

Cost of 3 metres of barbed wire
$$= 3b$$
Cost of 2 metres of plain wire
$$= 2w$$
$\Rightarrow$ Cost of 1 metre of fencing
$$= 3b + 2w$$
$\therefore$ Cost of $F$ metres of fencing
$$= (3b + 2w)F$$

4. Number and Algebra, NAP-89736

$$a = b - 3$$

5. Number and Algebra, NAP-25117

Consider each option:

$(m+n) \times l \ \rightarrow$ negative

$(m-n) \div l \ \rightarrow$ negative

$(m-n) \div -l \ \rightarrow$ postive

$(m+n) \times -l \ \rightarrow$ postive

Since $(m+n) > (m-n)$, and $-l$ is
is a positive whole number,

$\Rightarrow (m+n) \times -l$ gives the largest positive value.

6. Number and Algebra, NAP-42727

The extra ants in the 30 cm high anthill
$$= 100\ 000\ 000 - 10\ 000\ 000$$
$$= 90\ 000\ 000$$
$$= 90 \text{ million}$$

Worked Solutions